COMPTUNE ENGINEERING

HSE Pacific Auto-Stop Modifications Sample Machine: Komatsu WA470-8 SN:100804 PIN:KMTWA129KNA100804	Document Reference: B2600003A
Inspection Date:	01/07/2026
Inspection Location:	Veolia Chullora 75 Anzac Street, Chullora NSW 2190
Australian Standard AS 3450:2021 Earth-moving machinery - Wheeled or high-speed rubber-tracked machines - Performance requirements and test procedures for brake systems	

Background

Investigation into retro fitment of auto stop capabilities to machinery. Sample vehicle inspected Komatsu WA470-8. The intention of the device is to detect pedestrians that may be in the path of the vehicle and apply the service brakes of the vehicle to retard motion.



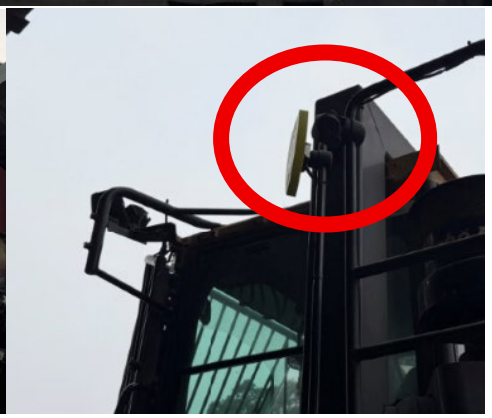


Inspection Findings:

Inspection of the sample vehicle was conducted on 01/07/2026. Vehicle had been retrofitted with an Auto Stop modification kit. The kit includes detection cameras, fitted to the rear of the vehicle and both sided facing rearward. This is to assist with detection of pedestrians which may be visible to the operator.







The braking system on the sample vehicle is hydraulically actuated to wet disc brakes which are actuated on four wheels.

BRAKES

Service brakes ..... Hydraulically actuated, wet disc brakes actuate on four wheels
Parking brake.....Wet disc brake
Emergency brake.....Parking brake is commonly used

A relay has been wired into the detection cameras and sends a signal to a solenoid that has been teed into the braking circuit after the accumulator . When a signal is sent, the service brakes are applied. A manual adjustment knob is at the end of the solenoid which allows for fluctuation of brake line flow within the maximum output from the accumulator in the new circuit that is established. This can be used to vary the brake line flow rate and can be adjusted based on the weight of the machine and the speeds commonly used. For example, higher speeds would need a higher line flow to stop in a timely manner. Slower speeds would only need a smaller line flow to make the ride quality better for the operator.



As this system is teed into the braking system beyond the accumulator for the service brakes, the function of the service remains. With the new circuit the original brake system remains unaltered and the set pressures for the service brakes remains the same OEM specifications, this braking application restricts flow at a less rate. For example, if a pedestrian is detected by a camera, brakes are applied and adjusted to say 50%-line flow, the operator can still apply the service brakes via the pedal and achieve a greater line flow if required.

After fitment of the system a test drive is required to verify functionality of the system and make adjustments to the adjustment knob on the solenoid as required.

Australian Standard AS 3450:2021 is the relevant standard for Earth-moving machinery - Wheeled or high-speed rubber-tracked machines - Performance requirements and test procedures for brake systems.

Summary:

Based on the inspected vehicle and the auto stop modification kit, this method of install and application could be used on machines with a similar braking system that is hydraulically actuated. The Australian standard AS3450:2021 has been reviewed against this modification. Care would need to be taken when fitting the kits and verification of their functionality must be carried out via a test drive after installation.

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WA470-8

SPECIFICATIONS

ENGINE

Model ..... Komatsu SAA6D125E-7*
Type ..... Water-cooled, 4-cycle
Aspiration ..... Turbo-charged, after-cooled, cooled EGR
Number of cylinders 6
Bore ..... 125 mm
Stroke ..... 150 mm
Piston displacement ..... 11.04 ltr
Governor ..... All-speed, electronic
Horsepower:
SAE J1995 ..... Gross 204 kW 273 HP
ISO 9249 / SAE J1349 ..... Net 203 kW 272 HP
Rated rpm ..... 2000 rpm
Fan drive method for radiator cooling ..... Hydraulic
Fuel system ..... Direct injection
Lubrication system:
Method ..... Gear pump, force-lubrication
Filter ..... Full-flow type
Air cleaner ..... Dry type with double elements and dust evacuator, plus dust indicator

*EPA Tier 4 Final emissions certified

TRANSMISSION

Torque converter 3-elements, 1-stage, 2-phase
Transmission ..... Automatic full-powershift, countershaft type

Travel speed	Forward*	Reverse*
1st	7.6 km/h	7.9 km/h
2nd	13.1 km/h (13.2 km/h)	13.5 km/h (13.7 km/h)
3rd	22.9 km/h (23.6 km/h)	23.6 km/h (24.3 km/h)
4th	36.2 km/h (38.3 km/h)	37.3 km/h (39.0 km/h)

*P-mode Measured with 26.5-R25 tyres ( ): Lock-up clutch ON

AXLES AND FINAL DRIVES

Drive system ..... Four-wheel drive
Front ..... Fixed, semi-floating
Rear ..... Centre-pin support, semi-floating, 26° total oscillation
Reduction gear ..... Spiral bevel gear
Differential gear ..... Conventional type
Final reduction gear ..... Planetary gear, single reduction

BRAKES

Service brakes ..... Hydraulically actuated, wet disc brakes actuate on four wheels
Parking brake ..... Wet disc brake
Emergency brake ..... Parking brake is commonly used

STEERING SYSTEM

Type ..... Articulated type, fully-hydraulic power steering
Steering angle ..... 35° each direction (40° to max end stop)
Minimum turning radius
at the centre of outside tyre ..... 6830 mm

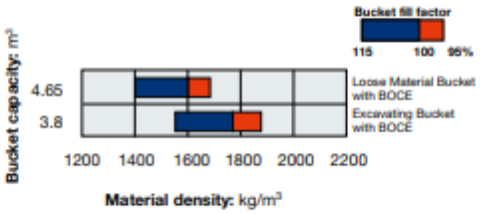
HYDRAULIC SYSTEM

Steering system:
Hydraulic pump ..... Piston type
Capacity ..... 195 ltr/min at rated rpm
Relief valve setting ..... 24.5 MPa 250 kgf/cm²
Hydraulic cylinders:
Type ..... Double-acting, piston type
Number of cylinders 2
Bore x stroke ..... 90 mm x 441 mm
Loader control:
Hydraulic pump ..... Piston pump
Capacity ..... 260 ltr/min at rated rpm
Relief valve setting ..... 34.3 MPa 350 kgf/cm²
Hydraulic cylinders:
Type ..... Double-acting, piston type
Number of cylinders—bore x stroke:
Lift cylinder ..... 2- 140 mm x 764 mm
Bucket cylinder ..... 1- 160 mm x 575 mm
Control valve ..... 2-spool type
Control positions:
Boom ..... Raise, hold, lower, and float
Bucket ..... Tilt-back, hold, and dump
Hydraulic cycle time (rated load in bucket)
Raise ..... 5.8 s
Dump ..... 1.6 s
Lower (Empty) ..... 3.1 s

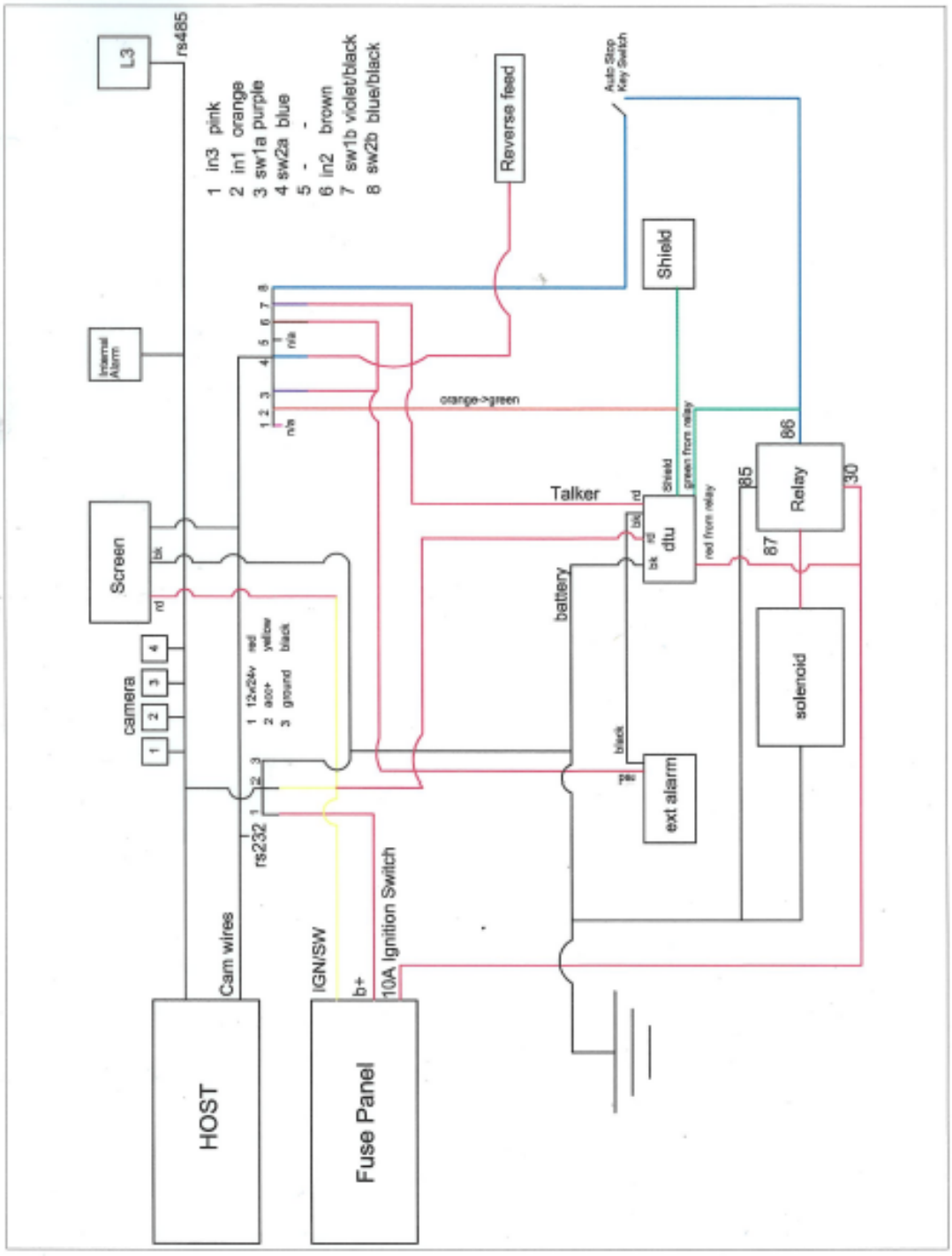
SERVICE REFILL CAPACITIES

Cooling system ..... 78 ltr
Fuel tank ..... 380 ltr
Engine ..... 38 ltr
Hydraulic system ..... 173 ltr
Axle front ..... 57 ltr
Axle ear ..... 56 ltr
Torque converter and transmission ..... 65 ltr
DEF tank ..... 36 ltr

BUCKET SELECTION GUIDE



Appendix B


L3
rs485
1 in3 pink
2 in1 orange
3 sw1a purple
4 sw2a blue
5 - -
6 in2 brown
7 sw1b violet/black
8 sw2b blue/black
Reverse feed
Auto Stop Key Switch
Shield
Internal Alarm
orange->green
Talker
Screen
camera
battery
dtu
Relay
86
85
87
30
solenoid
ext alarm
rs232
Cam wires
IGN/SW
b+
10A Ignition Switch
HOST
Fuse Panel